Technical Project:

# Enterprise Global WAN Architecture Design

Environment: Cisco Packet Tracer

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# Network Addressing and LAN Topology

## VLSM Design:

I used Variable Length Subnet masking (VLSM) for the table below. VLSM is a network design technique used to create IP networks with different subnet masks, it allows network administrators to assign larger subnet masks to subnets having more hosts and smaller subnet masks to the subnets having less hosts (GeekforGeeks, 2025). My allocation order is in descending order based on which zone had the most hosts to the least amount of hosts. This design preserves space for route summarisation by assigning addresses sequentially from 10.20.0.0 upwards placing two or more VLANs in the same summary route for example the corporate Wi-Fi and the operations zones can be advertised as 10.20.0.0/24 route since they occupy the whole address, this reduces the number of entries in the routing table. There are reserved continuous space in this design that allows for future growth, for example the address 10.20.2.0/24 is has 127 unassigned address that can be used to create new VLANs as the organization grows.

| Allocation Order | Zone | Total Hosts | Prefix | Network address | Usable Range | Broadcast Address | Gateway IP | VLAN ID |
|---|---|---|---|---|---|---|---|---|
| 1 | Corporate Wi-Fi | 120 | /25 | 10.20.0.0 | 10.20.0.1 – 10.20.0.126 | 10.20.0.127 | 10.20.0.1 | 50 |
| 2 | Operations | 100 | /25 | 10.20.0.128 | 10.20.0.129 - 10.20.0.254 | 10.20.0.255 | 10.20.0.129 | 30 |
| 3 | Finance | 80 | /25 | 10.20.1.0 | 10.20.1.1 - 10.20.1.126 | 10.20.1.127 | 10.20.1.1 | 10 |
| 4 | Guest Wi-Fi | 60 | /26 | 10.20.1.128 | 10.20.1.129 - 10.20.1.190 | 10.20.1.191 | 10.20.1.129 | 99 |
| 5 | HR | 50 | /26 | 10.20.1.192 | 10.20.1.193 - 10.20.1.254 | 10.20.1.255 | 10.20.1.193 | 20 |
| 6 | IT | 40 | /26 | 10.20.2.0 | 10.20.2.1 - 10.20.2.62 | 10.20.2.63 | 10.20.2.1 | 40 |
| 7 | Server Farm | 30 | /27 | 10.20.2.64 | 10.20.2.65 - 10.20.2.94 | 10.20.2.95 | 10.20.2.65 | 60 |
| 8 | Network Management | 20 | /27 | 10.20.2.96 | 10.20.2.97 - 10.20.2.126 | 10.20.2.127 | 10.20.2.97 | 70 |

## Three-Layer Architecture:

The LAN topology I propose for the Johannesburg HQ consists of three layers: core, distribution, and access. The access layer is made up of layer 2 switches that are connected directly to the end-user devices like workstations and printers. The ports connecting the switches to the device will be configured as access ports and will be assigned to specified VLANs to logically separate the Network. The segmentation will improve security and performance by isolating traffic in each zone. The distribution layer will be made up of a layer 3

switch that connect the core layer to the access layer. The switch will connect the layer 2 switches through a trunk link to allow the network from Multiple VLANs to use a single uplink. Inter-VLAN routing will be handle by the layer 3 Switch through Switch Virtual Interfaces (SVI). This will reduce the load on the core router and ensure that only traffic travelling to the external network is forwarded to the router. Two separate access points will be deployed for wireless connectivity, one for the corporate Wi-Fi and another for the guest Wi-Fi. The wireless access points will be connected to access ports on the layer 3 switch. The guest Wi-Fi will be isolated from accessing corporate resources using Access Control Lists (ACL), only allowing internet connection and restricting internal communication. This topology allows the organisation to scale  horizontally because new switches can be added to the access layer without depending on the limited number of router interface. Using the layer 3 switch for inter-VLAN routing enhances network performance and reliability, while ensuring efficient traffic management and a clear separation of roles.

### Wireless Security:

The two security measures I recommend for strengthening the wireless network are the following:

- WPA3-Enterprise with 802.1X authentication: WPA3 is the latest standard for Wi-Fi protected Access( WPA) developed by the Wi-Fi Alliance Security Technical Task Group. The enterprise version uses 802.1X authentication for certificate-based security with strong encryption (Cisco, 2025). This security measure requires users to authenticate themselves with unique credentials against a server. It helps reduce the risk of unauthorized Access in the corporate network unlike a pre-shared key (PSK) that can be guest. It also simplifies access management  because if an employee leaves, their credentials can be disabled on the serve revoking their access.
- Client Isolations on guest Wi-Fi: Client isolation is a feature that prevents wireless clients from communicating with one another on layer 2 as explain by (Cisco, 2025). This feature will help the risk of lateral movement between devices. This means that if a guest devices connected Wi-Fi is injected by malware it won’t be able to spread the malware to other guest devices.

# Global WAN Routing Strategy

## Hybrid Protocol Implementation:

The routing protocols that have the potential to meet TransGlobe Logistics grows requirements are OSPF and EIGRP.

- Open Shortest Path First (OSPF) is a link-state routing protocol and Interior Gateway Protocol (IGP) that uses a shortest Path First (SPF) algorithms to determine the shortest path to route packets in an IP network (IBM, 2025). Enhanced Interior Gateway Routing Protocol (EIGRP) is an advanced distance-vector routing protocol that also has characteristics of link-state protocols which allows for efficient routing within autonomous systems (Labs, 2025). Below are their differences:
    - Scalability: OSPF offers better scalability than EIGRP because it uses a hierarchical area structure to segment the network, allowing more areas to be added as the company grows. It limits the scope of link-state updates and reduces the size of the routing table of routers that are not in the backbone (area 0). While EIGRP has a single large domain that may require manual route summarization as the network grows.
    - Efficiency: OSPF floods its areas with Link State Advertisements(LSAs) during an initial convergence and when the network is updated to maintain convergence which consumes significant CPU cycles on routers and wastes link bandwidth. While EIGRP uses Diffusing Update Algorithm (DUAL) for rapid convergence and  to maintain a topology table by sends incremental updates only when there’re changes in the network to maintain convergence, this uses less bandwidth and CPU.
    - Policy control : EIGRP uses route maps to easily implement policies to have prise control over route filtering and summarization at any point in the network. While OSPF typically implements policies at Area Border Routers, primarily focusing on route filtering and summarization between areas.
    - High-Latency Links: EIGRP performs better than OSPF In these environments because it relies on reliable transport and send partial updates unlike  OSPF that send link-state databases across the network consuming more bandwidth.
    - Fault tolerance: EIGRP offers a better fault tolerance for large networks because it does not need a full recalculation for the backup route unlike OSPF that needs to recompute its SPF tree after a failure which takes longer in large networks and can lead to dropped packets.
    - Network Growth: OSPF make expanding easier because new region and branches  can be added as new areas without affecting the backbone. Since EIGRO is like a large single domain it makes it difficult to manage across multiple autonomous systems such as the one TransGlobe Logistics has.

## High-Latency Optimization

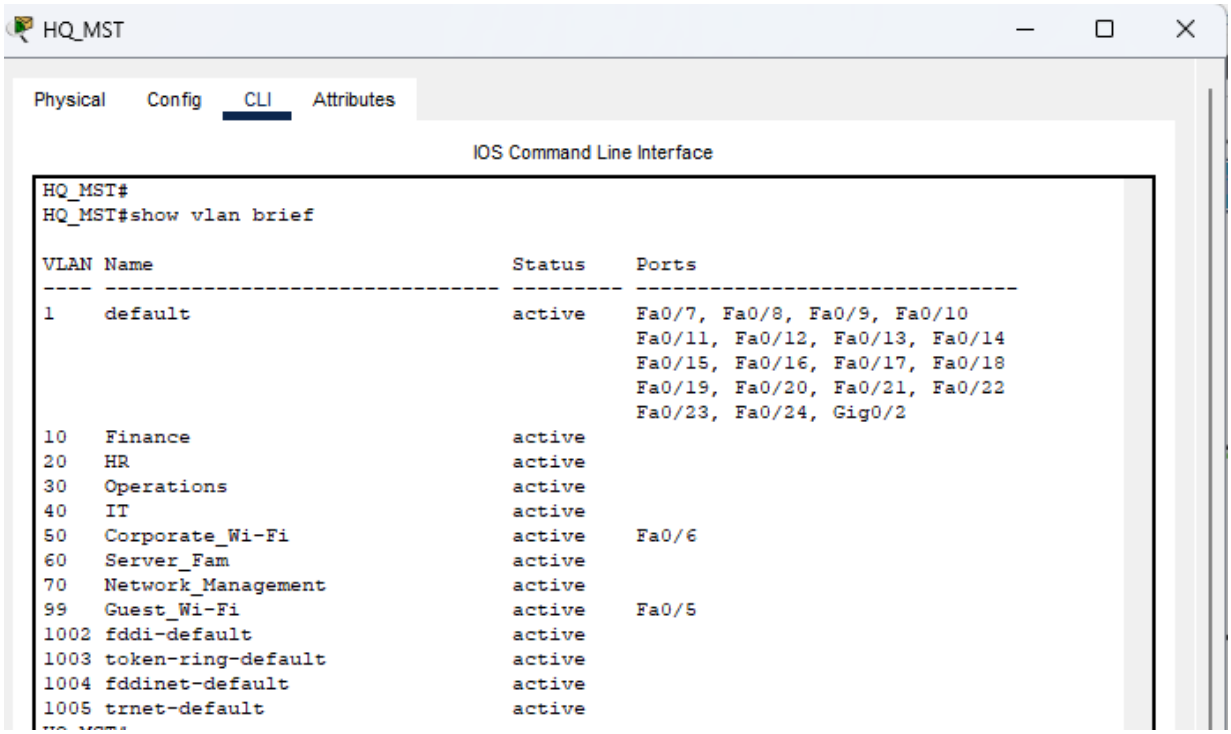
I recommend using a hybrid routing strategy that uses both EIGRP and OSPF across the Global WAN. EIGRP can be used for site-level routing and OSPF for inter-region routing. This design take advantage of the strengths of both the protocols for routing efficiency, scalability, and network resilience in this mixed fibre-and-satellite environment. At each regional site such as Johannesburg, London, and Singapore, EIGRP will be the internal routing protocol. Using DUAL

for fast convergence and sending incremental updates to minimize bandwidth consumption on the high-latency satellite links, ensuing that branch warehouses can maintain a stable connectivity and have quick failover even link conditions are not stable. OSPF will be used between regional routers so that we can use it’s hierarchical area design for scalable growth as new offices are added. The route summarisation at ABRs will be used to limit updates between regions and keep the routing tables smaller. Each regional router will perform redistribution between EIGRP and OSPF to integrate site-level and inter-region routing. This will allow local networks to only advertise summarized routes to the global backbone, which improves routing efficiency and prevents unnecessary satellite traffic.

# Implementation and Failover Validation

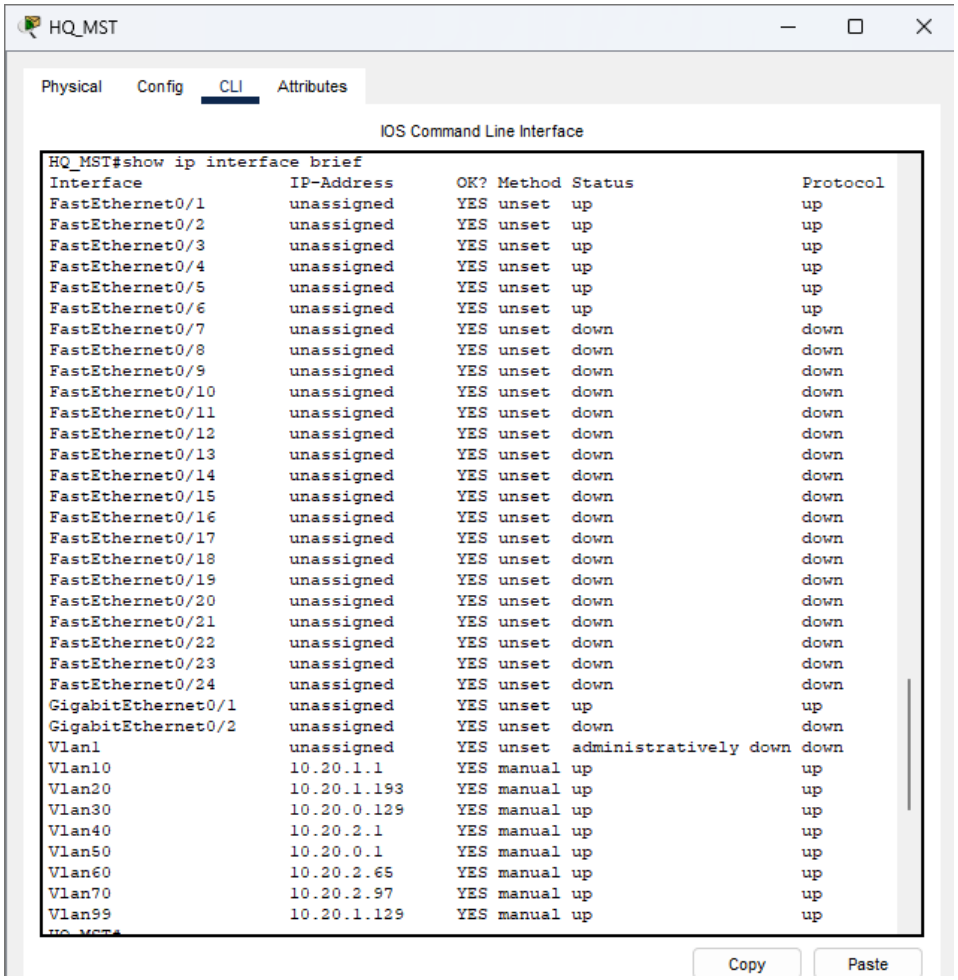
## Inter-VLAN Routing:

Results for show VLAN brief configured with the help of (IT, 2020):



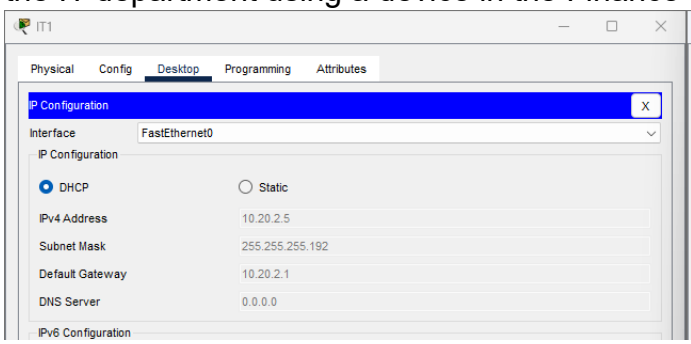

```
HQ_MST#
HQ_MST#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
10   Finance                          active
20   HR                               active
30   Operations                       active
40   IT                               active
50   Corporate_Wi-Fi                  active    Fa0/6
60   Server_Fam                       active
70   Network_Management               active
99   Guest_Wi-Fi                      active    Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

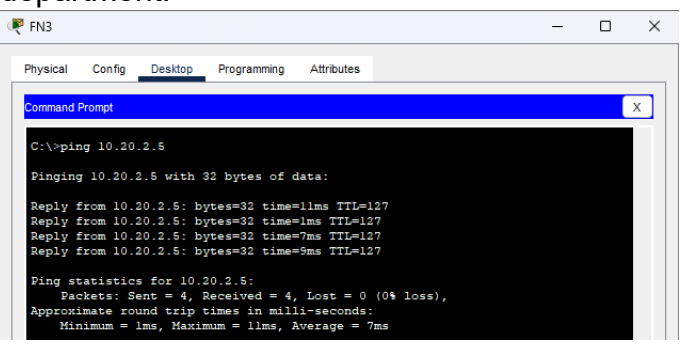
Results for show ip interface brief:



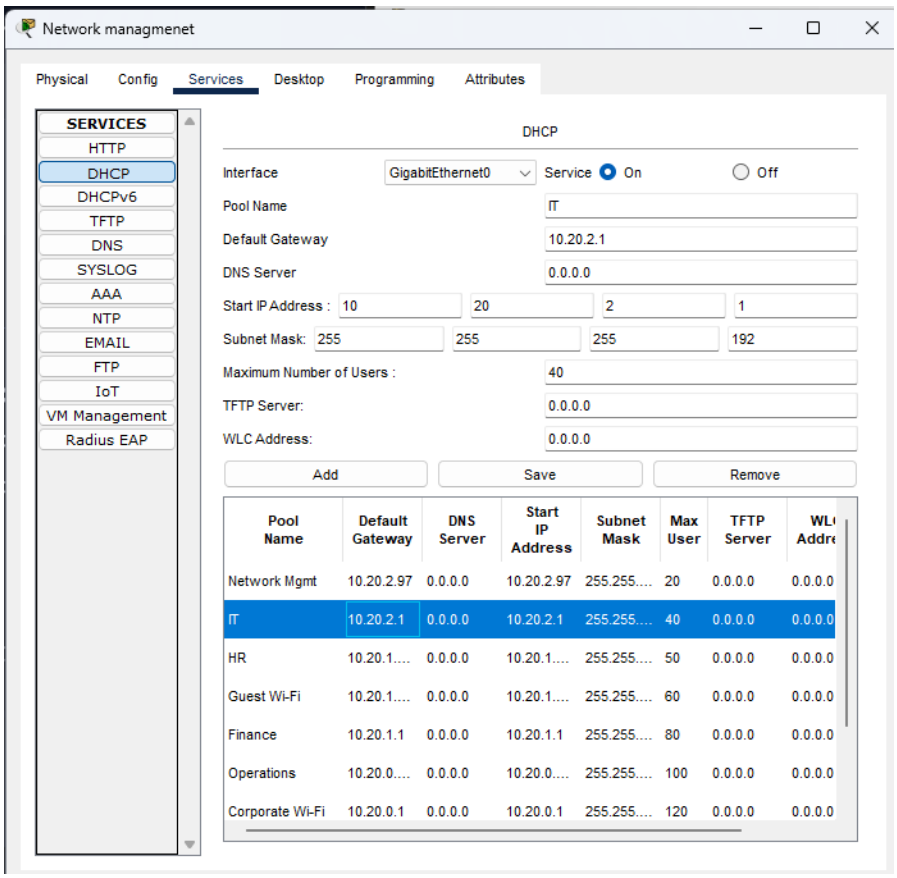

```
HQ_MST#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/2        unassigned      YES unset  up                    up
FastEthernet0/3        unassigned      YES unset  up                    up
FastEthernet0/4        unassigned      YES unset  up                    up
FastEthernet0/5        unassigned      YES unset  up                    up
FastEthernet0/6        unassigned      YES unset  up                    up
FastEthernet0/7        unassigned      YES unset  down                  down
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down
FastEthernet0/22       unassigned      YES unset  down                  down
FastEthernet0/23       unassigned      YES unset  down                  down
FastEthernet0/24       unassigned      YES unset  down                  down
GigabitEthernet0/1     unassigned      YES unset  up                    up
GigabitEthernet0/2     unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 10.20.1.1       YES manual up                    up
Vlan20                 10.20.1.193     YES manual up                    up
Vlan30                 10.20.0.129     YES manual up                    up
Vlan40                 10.20.2.1       YES manual up                    up
Vlan50                 10.20.0.1       YES manual up                    up
Vlan60                 10.20.2.65      YES manual up                    up
Vlan70                 10.20.2.97      YES manual up                    up
Vlan99                 10.20.1.129     YES manual up                    up
```

Prove that end device use DHCP and that Inter-vlan routing is possible by ping a device in the IT department using a device in the Finance department:

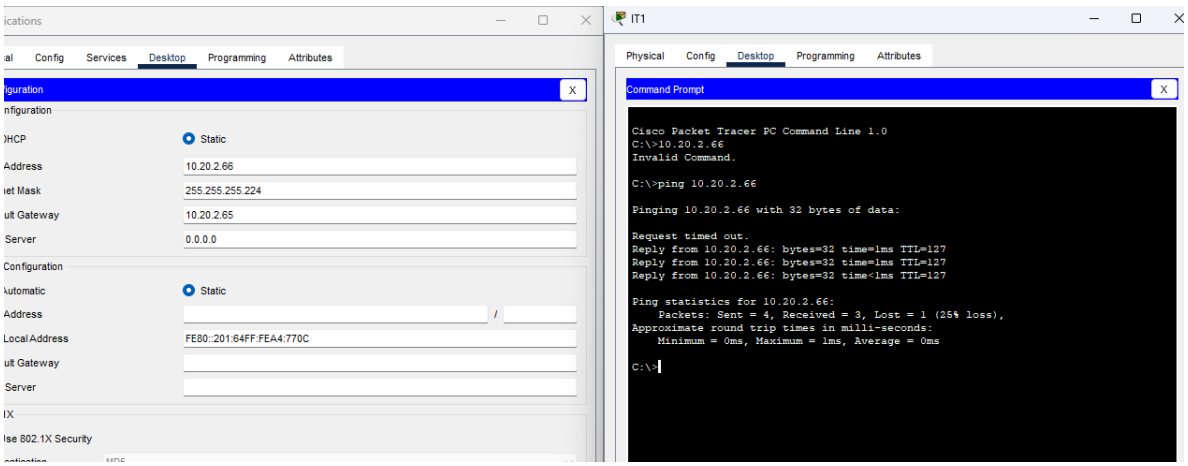



Prove that DHCP scopes have been well defined:




Prove that device can access server fam :




## Dynamic Failover Simulation:

Below is the prove that I'll implemented the hybrid module I've recommend of using OSPF as a WAN routing protocol and EIGRP for site-to-site routing configured with the help of (RAvish, 2024)

HQ_RT

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
  EIGRP maximum metric variance 1
Redistributing: eigrp 10, ospf 1
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
     10.255.0.1/32
     172.16.0.8/30
     10.20.3.0/30
     10.20.0.0/22
  Routing Information Sources:
    Gateway         Distance      Last Update
    172.16.0.10     90            128342
  Distance: internal 90 external 170


Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.255.0.1
  It is an autonomous system boundary router
  Redistributing External Routes from,
    eigrp 10
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    172.16.0.0 0.0.0.3 area 0
    172.16.0.4 0.0.0.3 area 0
    10.255.0.1 0.0.0.0 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.255.0.1           110      00:12:39
    10.255.0.2           110      00:13:51
    10.255.0.3           110      00:13:50
  Distance: (default is 110)

HQ_RT#
HQ_RT#
```

Copy | Paste

Top

Below is the result of traceroute showing that the entire network can communicate

RemotePC

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
C:\>ping 10.20.1.2

Pinging 10.20.1.2 with 32 bytes of data:

Reply from 10.20.1.2: bytes=32 time=1ms TTL=125
Reply from 10.20.1.2: bytes=32 time=2ms TTL=125
Reply from 10.20.1.2: bytes=32 time=24ms TTL=125
Reply from 10.20.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 10.20.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 24ms, Average = 7ms

C:\>ping 10.20.0.130

Pinging 10.20.0.130 with 32 bytes of data:

Reply from 10.20.0.130: bytes=32 time=10ms TTL=125
Reply from 10.20.0.130: bytes=32 time=4ms TTL=125
Reply from 10.20.0.130: bytes=32 time=12ms TTL=125
Reply from 10.20.0.130: bytes=32 time=20ms TTL=125

Ping statistics for 10.20.0.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 20ms, Average = 11ms

C:\>tracert 10.20.0.130

Tracing route to 10.20.0.130 over a maximum of 30 hops:

  1   0 ms      1 ms      0 ms      192.168.3.1
  2   0 ms      1 ms      0 ms      172.16.0.9
  3   0 ms      0 ms      1 ms      10.20.3.2
  4   11 ms     0 ms      0 ms      10.20.0.130

Trace complete.

C:\>
```

Top

OP1

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IPv4 Address: 10.20.0.130

Subnet Mask: 255.255.255.128

Default Gateway: 10.20.0.129

DNS Server: 0.0.0.0

IPv6 Configuration

Automatic | Static

IPv6 Address: / 

Link Local Address: FE80::202:16FF:FE46:D6B2

Default Gateway:

DNS Server:

802.1X

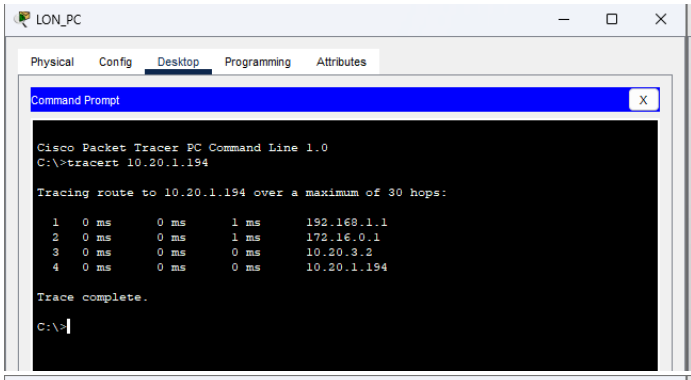
Use 802.1X Security

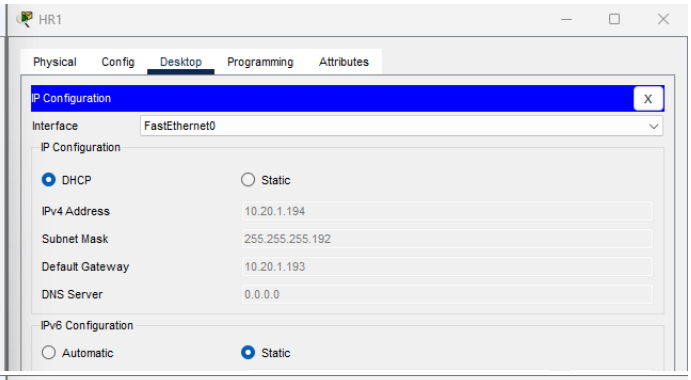
Authentication: MD5

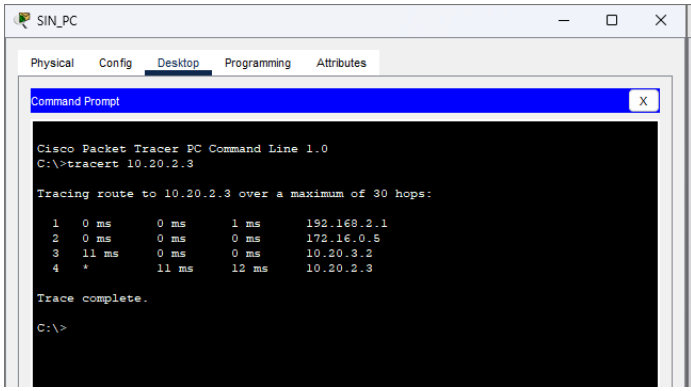
Username:

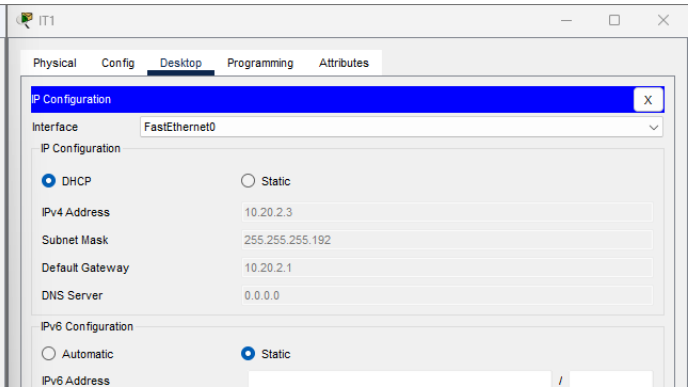
Password:

Top


LON_PC
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 10.20.1.194
Tracing route to 10.20.1.194 over a maximum of 30 hops:
1 0 ms 0 ms 1 ms 192.168.1.1
2 0 ms 0 ms 1 ms 172.16.0.1
3 0 ms 0 ms 0 ms 10.20.3.2
4 0 ms 0 ms 0 ms 10.20.1.194
Trace complete.
C:\>


HR1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.20.1.194
Subnet Mask
255.255.255.192
Default Gateway
10.20.1.193
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static


SIN_PC
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 10.20.2.3
Tracing route to 10.20.2.3 over a maximum of 30 hops:
1 0 ms 0 ms 1 ms 192.168.2.1
2 0 ms 0 ms 0 ms 172.16.0.5
3 11 ms 0 ms 0 ms 10.20.3.2
4 * 11 ms 12 ms 10.20.2.3
Trace complete.
C:\>


IT1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.20.2.3
Subnet Mask
255.255.255.192
Default Gateway
10.20.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address

Below is the route table of the HQ router before failure, it shows 10.255.0.3/32 being connected via 172.16.0.6 on GigabitEthernet1/0:

HQ_RT

Physical Config CLI Attributes

IOS Command Line Interface

```
HQ_RT#
HQ_RT#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
HQ_RT(config)#interface GigabitEthernet1/0
HQ_RT(config-if)#exit
HQ_RT(config)#exit
HQ_RT#
%SYS-5-CONFIG_I: Configured from console by console

HQ_RT#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
S       10.20.0.0/22 [4/0] via 10.20.3.2
C       10.20.3.0/30 is directly connected, GigabitEthernet0/0
C       10.255.0.1/32 is directly connected, Loopback0
O       10.255.0.2/32 [110/2] via 172.16.0.2, 00:19:56, GigabitEthernet2/0
O       10.255.0.3/32 [110/2] via 172.16.0.6, 00:02:11, GigabitEthernet1/0
D       10.255.0.4/32 [90/45248000] via 172.16.0.10, 00:18:48, Serial3/0
     172.16.0.0/30 is subnetted, 3 subnets
C       172.16.0.0 is directly connected, GigabitEthernet2/0
C       172.16.0.4 is directly connected, GigabitEthernet1/0
C       172.16.0.8 is directly connected, Serial3/0
O E2 192.168.1.0/24 [110/20] via 172.16.0.2, 00:19:56, GigabitEthernet2/0
O E2 192.168.2.0/24 [110/20] via 172.16.0.6, 00:02:11, GigabitEthernet1/0
D    192.168.3.0/24 [90/45122560] via 172.16.0.10, 00:18:48, Serial3/0

HQ_RT#
```

Copy Paste

Top

And here's the routing table after the link failure, it show  10.255.0.3/32 to be connected via 172.16.0.2 on GigabitEthernet2/0 which is the link between HQ and London.

HQ_RT

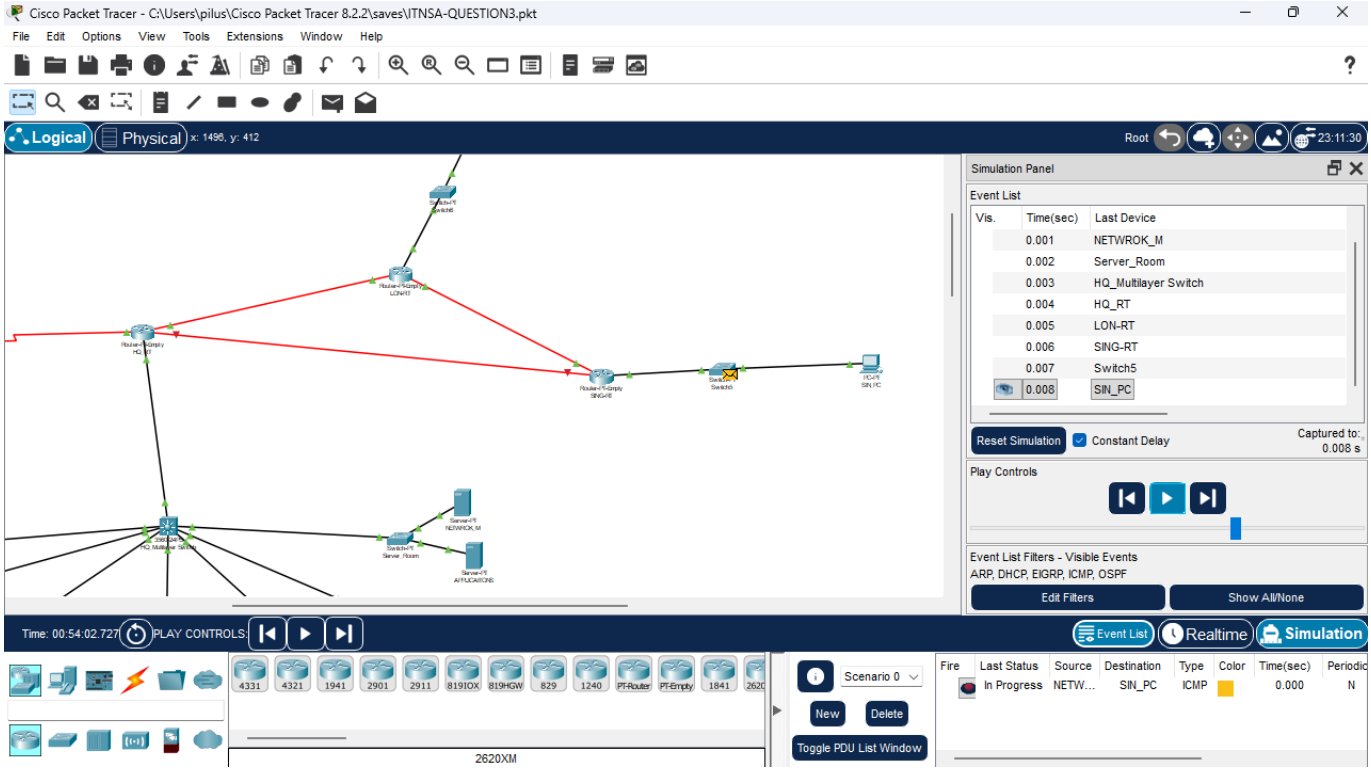
Physical Config CLI Attributes

IOS Comma

```
% Invalid input detected at '^' marker.

HQ_RT(config-if)#exxit
                  ^
% Invalid input detected at '^' marker.

HQ_RT(config-if)#exit
HQ_RT(config)#exit
HQ_RT#
%SYS-5-CONFIG_I: Configured from console by console

HQ_RT#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
S       10.20.0.0/22 [4/0] via 10.20.3.2
C       10.20.3.0/30 is directly connected, GigabitEthernet0/0
C       10.255.0.1/32 is directly connected, Loopback0
O       10.255.0.2/32 [110/2] via 172.16.0.2, 00:21:05, GigabitEthernet2/0
O       10.255.0.3/32 [110/3] via 172.16.0.2, 00:00:40, GigabitEthernet2/0
D       10.255.0.4/32 [90/45248000] via 172.16.0.10, 00:19:57, Serial3/0
     172.16.0.0/30 is subnetted, 2 subnets
C       172.16.0.0 is directly connected, GigabitEthernet2/0
C       172.16.0.8 is directly connected, Serial3/0
O E2 192.168.1.0/24 [110/20] via 172.16.0.2, 00:21:05, GigabitEthernet2/0
O E2 192.168.2.0/24 [110/20] via 172.16.0.2, 00:00:40, GigabitEthernet2/0
D    192.168.3.0/24 [90/45122560] via 172.16.0.10, 00:19:57, Serial3/0

HQ_RT#
HQ_RT#
```

Top

And here's a simulation that show's that the ICMP packet used the London route to reach the Singapore Offices:




The system achieved failover automatically due to the dynamic nature of OSPF, which forms a core backbone. IT also utilized the efficient update mechanism of EIGRP. When the link between HQ and Singapore failed, the OSPF adjacency dropped then the HQ router immediately detects the change and generate a new LSA that indicates that the link is down. The LSA flooded the backbone reaching London. London received the LSA and updated it's database, instantly running its SPF algorithm. The SPF calculation determines that the path from HQ to Singapore is through London and then All routers update their routing tables and traffic automatically flows though the redundant path via London.

# Performance Optimisation and Host Security

## Traffic Engineering:

The two techniques I recommend for optimizing performance across the WAN and LAN of TransGlobe Logistics are:

- Implementing Quality of service(QoS) across WAN routers: QoS is a technology used to manage network traffic by prioritizing important data to ensure good performance for critical applications. We are specifically going to use Congestion Management and DSCP (Differential Service Code point) an aspect of QoS used to classify and manage network traffic. Traffic for real-time application will be identified and marked with a high priority DSCP value such as EF-Expedited Forwarding or AF- Assured Forwarding at the HQ router interface closest to the application source. Traffic that is jitter intolerant is marked with the EF value which guarantees that the traffic will have low loss. Low latency and low jitter. This improves real-time service delivery. The traffic for large data updates will be marked with an AF call making sure it has priority assurance even if the link is congested (Cisco, 2025). Low Latency Queuing must also be configured on the congested satellite link to create a strict priority queue for the traffic marked with EF value. This will reduce jitter and delay while also improving the user experience.
- Caching: Caching is the process of storing copies of data in a temporary storage area to reduce the time between when the data is requested and when it is accessed (Sheldon, 2023). By deploying local cache servers, repetitive data that is frequently being accessed across the network can be copied and stored in the cache server so that the requests do have to go the central server. This reduces network congestion by having the data accessed locally and it also saves bandwidth on WAN links for traffic that can't be cashed. User experience is improved since users can excess the data faster improving work performance.

## Host-based Defence:

I recommend Endpoint Detection and Response (EDR) and Application whitelisting as host-based security controls for PCs and servers:

- EDR: According to (Fortinet, n.d.) EDR software is used by security operations teams to detect, contain, investigate and remediate cyberattacks such as ransomware and other malware. EDR software should be installed on all PCs and server to continuously monitor endpoint activity, record system events and analysis behaviour while communicating with centralized threat intelligence platforms. If a PCs is infected with malware, the software will detect the behaviour of the malware in real-time and automatically isolate the host from the network to prevent the malware from spreading across the network or discovering other device in the network. This reduces the risk of malware injections and lateral movement.
- Application whitelisting: Unlike blacklisting applications that are known to be malicious or unsafe which still leaves end devices vulnerable to running other malicious application that are not blacklisted. Application whitelisting is a security measure that only permits the execution of software applications or scripts that have been approved by management. All the other software that are not in the whitelist are blocked by default. This security measure reduces the risk of malware injection because most

modern viruses and ransomware rely on executing unauthorized code on the end device. So, by blocking all unapproved executables the malware won’t be able to run even if it bypasses other security controls within the network. This enhancing overall host protection and improve protection against zero-day threats.

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